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Bracketed Generic Inactivation of Rodent Retroviruses by Low pH Treatment for Monoclonal Antibodies and Recombinant Proteins

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Abstract: Viral safety is a predominant concern for monoclonal antibodies (mAbs) and other recombinant proteins (RPs) with pharmaceutical applications. Certain commercial purification modules, such as nanofiltration and low-pH inactivation, have been observed to reliably clear greater than 4 $\log_{10}$ of large enveloped viruses, including endogenous retrovirus. The concept of "bracketed generic clearance" has been proposed for these steps if it could be prospectively demonstrated that viral $\log_{10}$ reduction value (LRV) is not impacted by operating parameters that can vary, within a reasonable range, between commercial processes. In the case of low-pH inactivation, a common step in mAb purification processes employed after protein A affinity chromatography, these parameters would include pH, time and temperature of incubation, the content of salts, protein concentration, aggregates, impurities, model protein p*I*, and buffer composition. In this report, we define bracketed generic clearance conditions, using a prospectively defined bracket/matrix approach, where low-pH inactivation consistently achieves $\geq$4.6 $\log_{10}$ clearance of xenotropic murine leukemia virus (X-MLV), a model for rodent endogenous retrovirus. The mechanism of retrovirus inactivation by low-pH treatment was also investigated. © 2003 Wiley Periodicals, Inc. †*Biotechnol Bioeng* **82**: 321–329, 2003.
Keywords: viral clearance; monoclonal antibodies; recombinant proteins; low pH unit operations

INTRODUCTION

Viral safety is a predominant concern for monoclonal antibodies (mAbs) and other recombinant proteins (RPs) with pharmaceutical applications. Mammalian cell cultures that produce many of these products concomitantly produce endogenous retroviruses, and have on occasion been infected with adventitious viruses (Anderson et al., 1991; Bartal et al., 1982; Garnick, 1998; Lieber et al., 1973). Extensive cell bank and cell culture harvest screening for adventitious agents, quantification of endogenous retrovirus titers in cell culture harvests, and virus removal/inactivation studies of the purification process are integral components of product safety assurance strategies (Center for Biologics Evaluation and Research, 1997; International Conference for Harmonization, 1998). Virus removal/inactivation studies evaluate the clearance capacity of individual steps in the purification process to determine the $\log_{10}$ reduction value (LRV) for particular model viruses. The LRV of orthogonal steps (modules where clearance mechanisms differ) are added together to compute the overall process LRV for comparison to the quantity of endogenous retrovirus in the cell culture harvests.

Certain purification modules, such as nanofiltration and low-pH inactivation, have been observed to reliably clear >4 $\log_{10}$ of large enveloped viruses, including endogenous retrovirus (Center for Biologics Evaluation and Research, 1997). Because these modules are so robust between processes, it is now possible that the concept of "bracketed generic clearance" can be applied to them if supported by appropriate, prospectively generated data. A "generic clearance" validation study generates data using one model mAb or RP that can be applied to other protein, as long as the key parameters of the step are held constant between the model and the other mAbs or RPs (Center for Biologics Evaluation and Research, 1997). "Bracketing," or defining the ranges of these key parameters that still achieve acceptable outcomes, is permissible for product stability and virus clearance validation studies (Center for Biologics Evaluation and Research, 1997; Fairweather, 1995; International Conference for Harmonization, 2000) and potentially applicable to other studies, e.g., host cell protein and DNA removal. Thus, applying the bracketing concept to generic clearance raises the possibility of defining acceptable ranges of key

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parameters of particular steps like low pH that can be applied to almost all mAbs and RPs. The key requirement needed to justify bracketed generic clearance to viral clearance of particular steps would be a demonstration that LRV is not impacted by potential parameter differences, within a reasonable range, between processes. In the case of low-pH inactivation, a common step in mAb purification processes employed after protein A affinity chromatography, these parameters would include pH, time and temperature of incubation, salt content, protein concentration, aggregates, impurities, model protein p*I* and sequence, and buffer composition.

In this report, we define bracketed generic clearance conditions, using a prospectively defined bracket/matrix approach, where low-pH inactivation consistently achieves $\geq$4.6 $\log_{10}$ clearance of X-MLV, a model for rodent endogenous retrovirus. The mechanism of retrovirus inactivation by low-pH treatment was also investigated.

Table Ib. Matrix design of Study II.

	Buffer					
	100 m*M* acetate			25 m*M* citrate		
Protein (mg/mL)	1	10–15	30–40	1	10–15	30–40
NaCl (m*M*)						
0	mAb2[b]		RP2[a]	RP1[a]		mAb2[b]
250		RP1[a]			RP2[a]	
500	mAb2[a]		RP2[b]	RP1[b]		mAb2[a]

[a]Duplicate samples from these conditions were analyzed in infectivity assays at 15- and 60-min intervals in addition to 0- and 30-min intervals to establish inactivation kinetics. In addition, X-MuLV genome and RT activity Q-PCR assays were performed on samples from 0-, 15-, 30- and 60-min intervals from these conditions.

[b]Duplicate samples of these conditions were spiked with 20% CHO or SP2/0 cell-free supernatant to rule out protective effects of extraneous cell culture proteins, lipids, or nucleic acids. These samples were also analyzed using the X-MuLV genome and RT activity Q-PCR assays at a 30-min interval.

MATERIALS AND METHODS

Statistical Design of Bracket/Matrix Studies

Factorial design concepts initially proposed for product stability studies (Fairweather, 1995; International Conference for Harmonization, 2000) were applied to design two matrices, denoted as Study I and II (Tables Ia and Ib). The studies were designed to evaluate the robustness of low-pH inactivation of X-MLV at pH 3.8 ± 0.1 for $\geq$30 min. The impact of three qualitative and 2 quantitative factors was built into the matrix designs: protein type (2 mAbs and 2 RPs with p*I* values ranging approximately from 3 to 9, qualitative), buffer type (100 m*M* acetate or 25 m*M* citrate buffer, qualitative), temperature (14 or 25°C, qualitative), NaCl content (0–500 m*M*, quantitative), and protein concentration (1–40 mg/mL, quantitative). Study I used a model mAb (mAb1) and a 3-factor by 3-level full factorial matrix design. Study II used three other proteins (mAb2, RP1, and RP2) and a combination matrix/bracketing design. An assessment of the impact of an additional robustness parameter, impurity content, was built into both Study I and II by spiking select matrix samples with CHO (Chinese hamster ovary) or SP2/0 (murine myeloma) cell-free supernatants (1:4 v/v). Select Study I and II samples were analyzed at additional time points (T = 0, 15, 30, and 60 min) using infectivity assays, and X-MLV genome and reverse transcriptase quantitative-PCR assays in an attempt to gain insights into how low pH inactivates retrovirus and to see if any of the parameters evaluated in Study I or II impacted inactivation kinetics. Three aggregated mAb samples (mAb2, mAb3, and mAb4) were evaluated for the impact of protein aggregation on retrovirus inactivation by low-pH treatment in a third study.

Table Ia. Matrix design of Study I.

	Buffer					
	100 m*M* acetate			25 m*M* citrate		
Protein (mg/mL)	1	10–15	30–40	1	10–15	30–40
NaCl (m*M*)						
0	mAb1[a]	mAb1	mAb1	mAb1	mAb1	mAb1[a]
250	mAb1	mAb1[b]	mAb1	mAb1	mAb1[b]	mAb1
500	mAb1	mAb1	mAb1[a]	mAb1[a]	mAb1	mAb1

[a]Duplicate samples from these four conditions were analyzed in infectivity assays at 15- and 60-min intervals in addition to 0- and 30-min intervals to establish inactivation kinetics. In addition, X-MuLV genome and RT activity Q-PCR assays were performed on samples from 0-, 15-, 30- and 60-min intervals from these four conditions.

[b]Duplicate samples of these two conditions were spiked with 20% CHO or SP2/0 cell-free supernatant to rule out protective effects of extraneous cell culture proteins, lipids or nucleic acids. These samples were also analyzed using the X-MuLV genome and RT activity Q-PCR assays at 30-min interval.

Xenotropic Murine Leukemia Virus (X-MLV)

X-MLV was purchased from BioReliance (Rockville, MD) or prepared at Genentech as follows: X-MLV was grown in chronically infected mink lung cells. After incubation at 37°C for 7 days, the culture medium was collected, clarified, and concentrated by ultracentrifugation. The pelleted virus was resuspended in NTB buffer (150 m*M* NaCl, 10 m*M* Tris, and 0.1% BSA at pH 8.0). The resulting virus stock was aliquoted and stored at −60°C or below until use. The X-MLV virus stock used had a titer of approximately 10^7 $TCID_{50}$/mL.

Model Proteins

Model proteins mAb1, mAb2, mAb3, mAb4, mAb5, and RP1 were produced using proprietary commercial processes at Genentech, Inc. RP2 is bovine β-lactoglobulin A purchased from Sigma-Aldrich (St. Louis, MO). The p*I* values of the model proteins in Study I and II are approximately 9 (mAb1), 7 (mAb2), 5 (RP2), and 3 (RP1).

Low-pH Incubations

All samples were incubated at pH 3.8 ± 0.1 for 30 min after a 1:10 (v/v) addition of X-MLV. The pH of each sample was verified after X-MLV addition using a pH meter. Incubation temperatures of samples in Study I (25 ± 1°C) and Study II (14 ± 1°C) were monitored using a thermometer. After 30 min, samples were neutralized by the addition of a pre-determined volume of 3 *M* Tris (pH 8.3). Duplicate samples in inactivation kinetic experiments were held at low pH for 60 min, with aliquots drawn and neutralized at intervals of 0, 15, 30, and 60 min. Samples from impurity interference experiments were spiked with either 20% (v/v) CHO cell-free supernatant (Genentech, Inc.) or 20% (v/v) SP2/0 cell-free supernatant (CBER/FDA) prior to low-pH treatment.

Spiked Impurities

CHO cell-free supernatants were collected from CHO cells grown at 12,000-L scale commercial production runs reaching terminal cell densities of ~5 × 10^6 cells/mL. SP2/0 cell-free supernatants were collected from SP2/0 cells grown in T-flasks containing serum-free media (Hybridoma SFM, Life Technologies, Gaithersburg MD). The SP2/0 cultures grew for 3 days, reaching a terminal cell density of ~0.3 × 10^6 cells/mL.

Aggregate Analysis

High-performance size-exclusion chromatography (HPSEC) was used for aggregate quantification in aggregated samples of mAb2, mAb3, and mAb4. HPSEC separates molecules according to their hydrodynamic volume with an elution order of decreasing molecular size. Therefore, HPSEC allows for the quantitative determination of the mAb purity as it relates to molecular size distribution by separating mAb monomer from aggregate, low molecular weight fragments, and excipients.

X-MLV Cell-Based Infectivity Assay

X-MLV titer was determined by median tissue culture infectious dose 50 ($TCID_{50}$) assay. Briefly, PG-4 (feline S^+L^-) cells were seeded into 96-well plates at a density of approximately 6 × 10^4 cells/mL. The plates were incubated at 36–38°C until the culture was 80–90% confluent. A serial dilution of the sample was made, and 50 μL of each dilution was inoculated into each of 8 replicate wells. The wells were examined for virally introduced cytopathic effect (CPE) following incubation at 36–38°C for 7 days. The virus titer was calculated using the Karber formula (Payment and Trudel, 1993).

Appropriate controls were included and assayed by the same methods. All samples were tested for cytotoxicity and interference with X-MLV infectivity. For cytotoxicity assessment, test samples at neutral pH without virus spiking were serially diluted and 50 μL of each dilution was inoculated into each of 8 replicate wells. Culture medium was inoculated on 8 replicate wells to serve as cytotoxicity negative control. For viral interference assessment, test samples at neutral pH with virus spiking were serial diluted and subsequently inoculated into 8 replicate wells per dilution at 50 μL/well. To serve as the positive control, a titration of the same X-MLV stock was also performed using culture medium for serial dilution. An effect of interference with X-MLV infectivity is defined as a change in virus titer of $\geq 10^{1.0}$ in the test sample treated titer over the titer for the positive control. All test samples were assayed at the lowest dilution shown to have no cytotoxicity and no interference with X-MLV infectivity.

When no virus was detected in the 30-min incubation samples using above 8 replicate wells/dilution standard format, the test sample was inoculated onto 80 replicate wells at 50 μL/well at the lowest sample dilution shown to have no cytotoxicity and no interference with X-MLV infectivity. Such larger testing volume increased assay sensitivity. When no virus was detected, the theoretical titer, i.e., the minimum assay detection limit, was calculated based on the Poisson distribution as follows: $c = -\ln(0.05)/v = 3/v$, where v is the total volume of the sample tested. This defines the concentration of virus that would have to be present in the sample in order to achieve a positive result in the fraction tested with 95% confidence.

TaqMan Assays

The quantitative PCR (Q-PCR) assay used to measure X-MLV reverse transcriptase (RT) activity was performed as previously described (Brorson et al., 2001). The Q-PCR assay used to quantify X-MLV genomic particle RNA copy number was performed as follows: total RNA was extracted from test samples using QIAamp Viral RNA kit (QIAgen Inc., Chatsworth, CA), according to the manufacturer's protocol. The Q-PCR reactions were carried out with 20 μL of the total RNA extracted from each test sample, 100 n*M* of forward primer, 100 n*M* of reverse primer, 100 n*M* of the probe, 6.5 m*M* of $MgCl_2$, 1.25 units of AmpliTaq Gold DNA polymerase, and 12.5 units of multiscribe MLV reverse transcriptase (Applied Biosystems, Foster City, CA) in a total volume of 50 μL. X-MLV particle RNA was first reverse transcribed into cDNA by incubation at 48°C for 30 min. Amplification began with a 10-min incubation at 95°C to activate AmpliTaq Gold DNA polymerase followed by 40 cycles of denaturation at 95°C for 15 sec, annealing and extension at 60°C for 60 s. The Q-PCR primers and probe used were as follows: forward primer 5′-GGC AGG AGC CTC GGT ACA A-3′, reverse primer 5′-TGT CAT TAG GTT GGT AAC TCT CCA AGT-3′, and the probe 5′-TGA CAG CCC TCA CCA GGT CTT CAA TG -3′. The probe was labeled with a fluorescent reporter dye FAM (6-

carboxyfluorescein) at the 5′ end and a quencher dye TAMRA (6-carboxytetramethylrhodamine) at the 3′ end. This primer set defines a 76-base pair (bp) fragment of the X-MLV envelope gene.

Thin-Section Transmission Electron Microscopy (TEM)

Low pH X-MLV samples were incubated for 1 h in 100 m*M* sodium acetate buffer at pH 3.8 containing 0, 250, or 500 m*M* NaCl. The virus solution was then neutralized to pH 7.0 by dropwise addition of 1 *M* Tris (pH 9.8). Proper control of the pH of all samples was monitored using a pH meter. Control virus was incubated at pH 7.0 in 250 m*M* NaCl. X-MLV in these samples was pelleted using high-speed ultracentrifugation (100,000*g*). The X-MLV pellets were fixed by a 10-min incubation with a glutaraldehyde solution (2% in PBS). Thin-section negative-stained transmission electron microscopy (150,000× magnification) of the fixed pellets was performed by JFE Enterprises (Brookeville, MD).

RESULTS

Study I

Study I was designed to evaluate the impact of protein concentration, impurities, buffer system, and NaCl concentration on retrovirus inactivation by 30 min of incubation at pH 3.8 (Table Ia). Study I was conducted using one mAb (mAb1, p*I* ~ 9) at 25 ± 1°C (a common commercial low-pH incubation temperature). LRV in each point of the study matrix was 4.6 or greater for all tested protein concentrations, NaCl concentrations, buffer systems, and impurities (Table IIa). Nearly complete inactivation was observed after 15 min of incubation in each of 4 conditions tested (Fig. 1a), while spiking samples with impurities had little impact on LRV. In some instances, infectious X-MLV was undetectable after low-pH inactivation. In these cases, data in Table II are expressed as ≥ the lowest possible LRV. Thus, a 30-min incubation in either citrate or acetate buffer at pH 3.8 inactivates ≥4.6 $\log_{10}$ of X-MLV.

Study II

Study II was designed to verify the conclusions drawn from Study I and to see if the conclusions apply to other model proteins (Table Ib). Study II was conducted using one mAb (mAb2, p*I* ~ 7) and two RPs (RP1, p*I* ~ 3; RP2, p*I* ~ 5) at 14 ± 1°C, the lowest temperature generally considered to be "room temperature". LRV at 30 min was ≥4.6 for each condition tested (Table IIb). The kinetics of inactivation at 14°C appears to be slower than at 25°C, as two conditions had one of the duplicate samples with an LRV < 4.0 $\log_{10}$ at 15 min (Fig. 1b). An apparent drop-off of LRV at 60 min

Table IIa. LRV[a] of X-MuLV infectivity after 30-min incubation at pH 3.8 ± 0.1 from Study I.

	Buffer					
	100 m*M* acetate			25 m*M* citrate		
Protein (mg/mL)	1	10–15	30–40	1	10–15	30–40
NaCl (m*M*)						
0	5.4 ± 0.9 ≥5.1 ± 0.3	≥5.6 ± 0.4	5.3 ± 0.6	5.4 ± 0.7	5.7 ± 0.7	5.0 ± 0.5 ≥5.4 ± 0.4
250	5.9 ± 1.0	≥5.0[b] ± 0.4 5.7[c] ± 0.8	5.3 ± 0.6	5.6 ± 0.7	4.9[b] ± 0.9 5.3[c] ± 0.6	4.6 ± 0.5
500	≥5.6 ± 0.4	5.4 ± 0.7	≥5.4 ± 0.4 5.1 ± 0.7	5.6 ± 0.7 5.0 ± 0.6	5.8 ± 1.0	5.0 ± 0.5

[a]Reported with 95% confidence interval.
[b]Sample was spiked with 20% CHO cell-free supernatant.
[c]Sample was spiked with 20% SP2/0 cell-free supernatant.

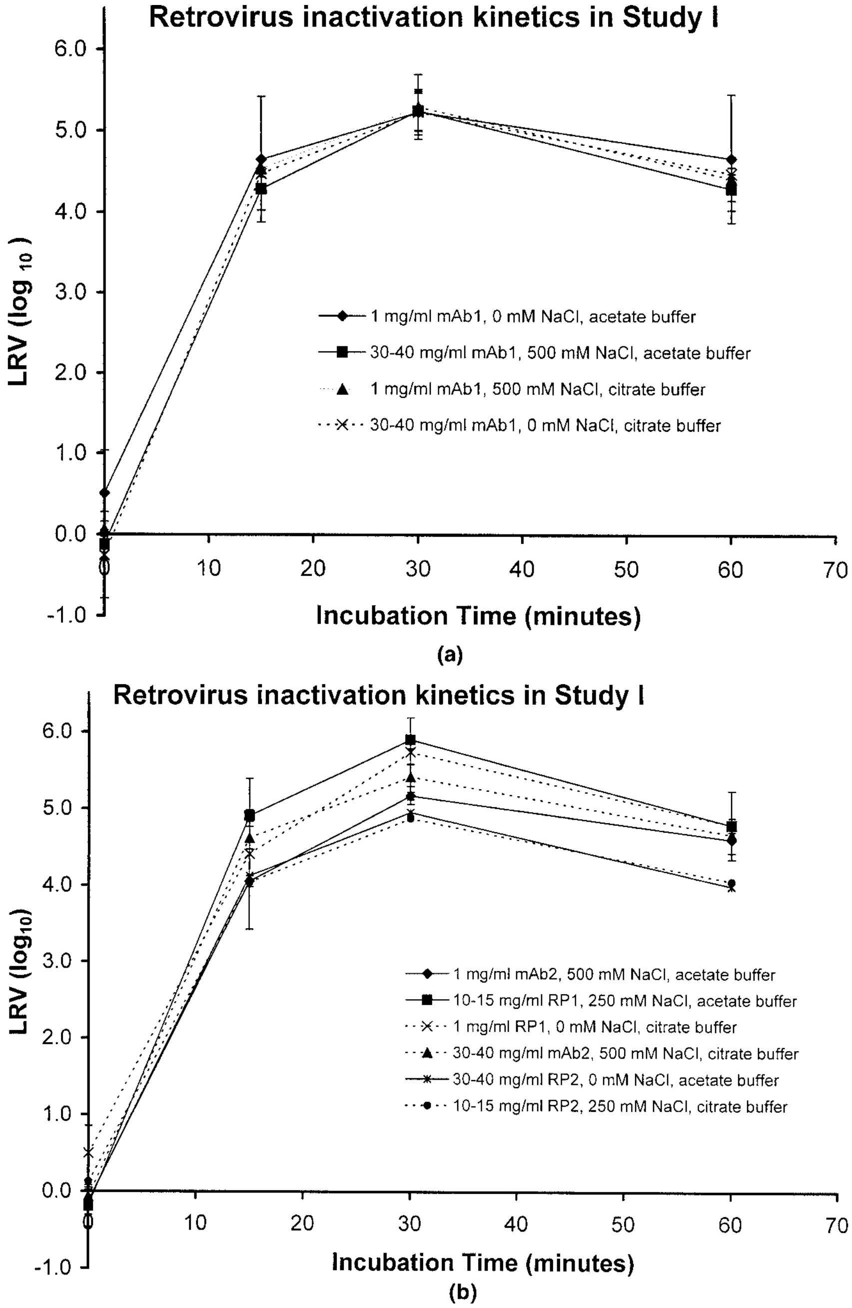

Figure 1. Kinetics of X-MLV inactivation during 60 min of incubation at pH 3.8 in Study I (**a**) and II (**b**). Inactivation conditions are indicated in the graph legend. Error bars are standard deviations of duplicate samples.

relative to 30 min is probably due to a more sensitive infectivity assay being used to measure X-MLV in the 30-min samples. In the 60-min samples, 8 wells were screened in the infectivity assay. Many wells contained no foci, resulting in LRV calculations that represented minimal estimates. In the 30-min samples, 80 wells, not 8, were screened in the infectivity assays, allowing a measurement of a higher LRV. Spiking samples with impurities had little impact on LRV in Study II. Thus, Study I and II together demonstrate that a 30-min incubation at pH 3.8 in either citrate or acetate buffer inactivates $\geq$4.6 $\log_{10}$ of X-MLV independently of model proteins (mAbs or RPs) as long as the temperature is $\geq$14°C, protein concentration is $\leq$40 mg/mL and NaCl concentration is $\leq$500 m*M*.

Study III (Aggregated Samples)

Low-pH conditions often induce denaturation and subsequent aggregation of proteins. To rule out the potential of protein aggregation impacting X-MLV inactivation, low-pH incubations were performed in the presence of highly aggregated mAbs (mAb2, 5%; mAb3, 18%; mAb4, 39%; Table III). In two of the three experiments (mAb2 and mAb4), >5 $\log_{10}$ of X-MLV inactivation occurred within 30

Table IIb. LRV[a] of X-MuLV infectivity after 30-min incubation at pH 3.8 ± 0.1 from Study II.

	Buffer											
	100 m*M* acetate						25 m*M* citrate					
Protein (mg/mL)	1		10–15		30–40		1		10–15		30–40	
NaCl (m*M*)												
0	≥5.7[b] ± 0.4	5.7[c] ± 0.9			5.3 ± 0.9	≥4.6 ± 0.4	≥5.4 ± 0.4	≥6.1 ± 0.5			5.7[b] ± 0.9	5.7[c] ± 0.9
250			≥5.7 ± 0.3	6.1 ± 1.0					4.9 ± 0.8	4.9 ± 0.7		
500	5.1 ± 0.6	5.2 ± 0.5			5.2[b,d] ± 1.0	5.2[c,e] ± 1.0	5.6[b] ± 0.5	5.3[c] ± 0.5			5.2 ± 0.6	≥5.7 ± 0.4

[a]Reported with 95% confidence interval.
[b]Sample was spiked with 20% CHO cell-free supernatant.
[c]Sample was spiked with 20% SP2/0 cell-free supernatant.
[d]Experiment performed with 18 mg/mL RP2 because of technical difficulties during RP2 sample concentration in presence of 500 m*M* NaCl. Experiment performed with 38 mg/mL mAb5 under same set of conditions resulted in an infectivity LRV of 5.3 ± 0.6 (20% CHO) and 5.6 ± 0.7 (20% SP2/0) at 30 min.

min. In the mAb3 experiment, LRV at 30 min was 4.3 $\log_{10}$, but the slightly lower LRV may have resulted from the buffer system (50 m*M* acetate, 500 m*M* Na_2SO_4) rather than an impact of aggregates. In the same experiment, LRVs of 2.5 and 5.3 were obtained at 15 and 60 min, arguing that there was a slower LRV kinetic for this sample. Thus, these data argue that low-pH inactivation performed in high concentrations of salts other than NaCl may need incubation times longer than 30 min. These data further argue that aggregates do not impact X-MLV inactivation by low pH.

Inactivation Mechanism

To explore how low pH inactivates retroviruses, retroviral RNA genomes and RT activity were quantified in the kinetic- and impurity-analysis samples using specific Q-PCR assays (Table IV). In each sample, the number of genome copies and level of RT activity after 30-min low-pH incubation was within ~1 $\log_{10}$ of the initial level, while infectivity was reduced by $\geq$4.6 $\log_{10}$. Thus, the near-complete loss of X-MLV infectivity after low-pH inactivation does not appear to involve destruction of either viral RNA genomes or RT molecules.

To ascertain if visible structural changes occur, thin-section TEM micrographs of X-MLV were taken without and after incubation for 1 h at pH 3.8 in acetate buffer containing 0, 250, and 500 m*M* NaCl (Fig. 2). Control virus, held at pH 7.0 in 250 m*M* NaCl, has well-defined, geometrically shaped capsids and clearly visible envelopes. Upon close inspection of the micrographs, rough outlines of envelope proteins can be seen. In contrast, viruses treated at pH 3.8 appear to have undergone some visible changes. Virus treated in the presence of NaCl still has distinguishable capsids and envelopes, but the capsids are darker and more mottled while some envelopes have stream-like projections. Viruses treated in the absence of NaCl are clearly more disrupted; the capsids are blurry and dark with little apparent membrane structure. Significant virus aggregation is also evident. However, ultracentrifugation over sucrose density gradients did not reveal density changes in X-MLV following low-pH treatment. The buoyant density of X-MLV before and after low-pH treatment was 1.17–1.18 g/mL (data not shown).

DISCUSSION

In this report, we demonstrate that low-pH incubation is a robust inactivation step for X-MLV, a virus used in virus clearance validation studies as a model for rodent type C retrovirus particles. We find that a 30-min incubation of X-MLV at pH 3.8 in either citrate or acetate buffer results in robust inactivation ($\geq$4.6 $\log_{10}$) when the protein concentration of any of four model proteins is $\leq$40 mg/mL and the NaCl concentration is $\leq$500 m*M*. The four model proteins included 2 mAbs and 2 RPs, with p*I* values ranging from approximately 3 to 9. Addition of cell culture derived

Table III. LRV of X-MuLV infectivity and genomes after 30-min incubation at pH 3.8 ± 0.1 in the presence of aggregated mAb.

mAb[a]	Buffer	Protein (mg/ml)	NaCl (m*M*)	Aggregate level (%)	LRV: Infectivity ($\log_{10}$ $TCID_{50}$)[d]	LRV: Genomes ($\log_{10}$ copies)[e]
mAb2	25 m*M* Tris[b]	0.3	500	5	5.1 ± 0.5	0.1 ± 0.0
mAb3	50 m*M* acetate	0.7	0[c]	18	4.3[c] ± 0.4	0.1 ± 0.0
mAb4	100 m*M* acetate	0.6	0	39	≥5.2 ± 0.5	0.1 ± 0.1

[a]X-MuLV was inactivated by incubation for 30 min at pH 3.8 at room temperature. The degree of aggregation of the model proteins was determined before low pH treatment using high-performance size-exclusion chromatography.
[b]Mab2, formulated in 25 m*M* Tris, pH 8.0, was acidified to pH 3.8 by addition of 1 *N* HCl.
[c]Low pH incubation was performed in the presence of 500 m*M* Na_2SO_4 instead of NaCl. Infectivity LRV at 15 and 60 min was 2.5 $\log_{10}$ and 5.3 $\log_{10}$, respectively.
[d]Reported with 95% confidence interval.
[e]Reported with standard deviation.

impurities or extreme aggregation of the model proteins did not inhibit low-pH inactivation. The kinetics of inactivation may be slightly slower at 14°C than at 25°C, but an LRV of ≥4.6 $\log_{10}$ is achieved at both temperatures by 30 min.

Based on these data, we propose that a "bracketed generic clearance" LRV of 4.6 $\log_{10}$ of rodent type C retrovirus can be applied to mAbs and RPs (CHO cell and murine hybridoma derived) that have recovery processes which include low-pH steps that meet the following conditions:

- pH is ≤3.8.
- Incubation time is ≥30 min.
- The incubation temperature is ≥14°C.
- The buffer system is acetate or citrate.
- Total protein concentration is ≤40 mg/mL.
- NaCl concentration is ≤500 m*M*.
- The p*I* of the mAb or RP is between approximately 3 and 9.
- The low-pH incubation step is performed on a cell-free intermediate after the initial capture chromatography step of the recovery process.
- The product is not a retrovirus targeted mAb.

We propose that this bracketed generic clearance LRV of 4.6 $\log_{10}$ can be substituted for product specific validation data if the low-pH step meets these conditions.

Our TEM studies have indicated that low-pH incubation impacts capsid and membrane structure of X-MLV in a manner sufficiently extensive that it can be visualized as membrane and capsid blurring and virus aggregation. In contrast, our Q-PCR data show that genomic RNA and RT activity are intact following low-pH exposure.

All of these data suggest that the mechanism of inactivation of murine retrovirus at pH 3.8 may possess similarities to low-pH inactivation mechanisms observed in other viruses. In these viruses, endosomal trafficking is part of their

Table IV. Comparison of LRV after 30-min incubation at pH 3.8 measured using infectivity and Q-PCR assays.

Study no.	Condition: Buffer	Condition: Protein (mg/mL)	Condition: NaCl (m*M*)	LRV: Infectivity[b] ($\log_{10}$ $TCID_{50}$)	LRV: Genomes[c] ($\log_{10}$ copies)	LRV: RT activity[c] ($\log_{10}$ picounits)
I	Acetate	1	0	5.4 ± 0.9, ≥5.1 ± 0.3	0.1 ± 0.5	1.0 ± 0.3
		10–15[a]	250	5.0 ± 0.4, 5.7 ± 0.8	−0.2 ± 0.1	−0.2 ± 0.2
		30–40	500	≥5.4 ± 0.4, 5.1 ± 0.7	0.0 ± 0.0	1.0 ± 0.4
	Citrate	1	500	5.6 ± 0.7, 5.0 ± 0.6	−0.1 ± 0.2	0.2 ± 0.2
		10–15[a]	250	4.9 ± 0.9, 5.3 ± 0.6	−0.0 ± 0.1	0.2 ± 0.1
		30–40	0	5.0 ± 0.5, ≥5.4 ± 0.4	0.1 ± 0.2	1.1 ± 0.9
II	Acetate	1	500	5.1 ± 0.6, 5.2 ± 0.5	0.1 ± 0.2	0.6 ± 1.6
		10–15	250	≥5.7 ± 0.3, 6.1 ± 1.0	0.1 ± 0.1	0.0 ± 1.0
		30–40	0	5.3 ± 0.9, ≥4.6 ± 0.4	0.9 ± 0.0	0.3 ± 0.6
		1[a]	0	≥5.7 ± 0.4, 5.7 ± 0.9	0.2 ± 0.2	1.2 ± 0.7
		30–40[a]	500	5.2 ± 1.0, 5.2 ± 1.0	0.7 ± 0.2	1.1 ± 0.2
	Citrate	1	0	≥5.4 ± 0.4, ≥6.1 ± 0.5	1.2 ± 0.0	0.6 ± 0.2
		10–15	250	4.9 ± 0.8, 4.9 ± 0.8	0.5 ± 0.3	−0.3 ± 1.5
		30–40	500	5.2 ± 0.6, ≥5.7 ± 0.4	0.2 ± 0.2	0.3 ± 0.4
		1[a]	500	5.6 ± 0.5, 5.3 ± 0.5	−0.1 ± 0.1	0.9 ± 1.2
		30–40[a]	0	5.7 ± 1.0, 5.7 ± 1.0	0.0 ± 0.1	0.5 ± 0.4

[a]Model protein at stated concentration also contains proteins from CHO or SP2/0 cell-free supernatants.
[b]Reported with 95% confidence interval.
[c]Average value from duplicate experiments, reported with standard deviation.

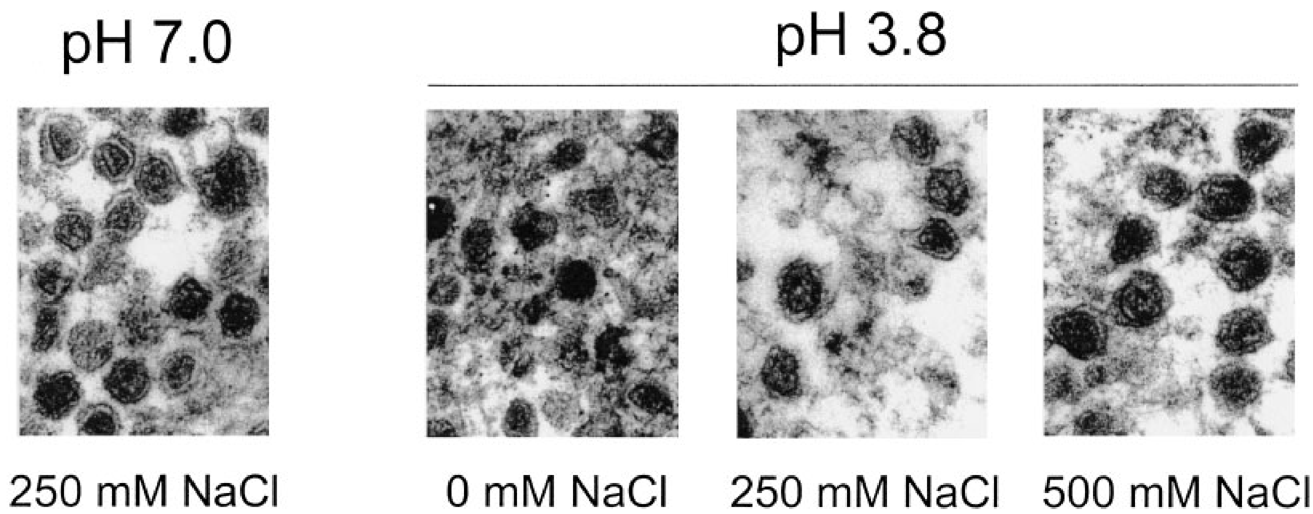

Figure 2. Thin-section transmission electron microscopy of control and low-pH inactivated X-MLV. Representative sections were selected from micrographs taken at 150,000× magnification. All samples were incubated at 20°C for 1 h in the presence of 1 mg/mL mAb5 at the indicated pH and NaCl content.

infection process, but incubation in mild acidic conditions similar to endosomes (pH ~ 5.0–6.0) was found to induce surface conformational changes that interfered with infectivity and membrane fusion (Bron et al., 1993; Corver et al., 2000; Gaudin et al., 1993; Gollins and Porterfield, 1986; Grgacic and Schaller, 2000; Kimura and Ohyama, 1988; Sato et al., 1983; Skehel et al., 1982; White and Wilson, 1987). Surface glycoprotein changes were detectable using methods such as protease digest pattern analysis, binding analysis using anti-peptide antibodies, transmission electron microscopy (TEM), and X-ray crystallography (Bullough et al., 1994; Carr and Kim, 1993; Huang et al., 1981; Scholtissek, 1985; Wilson et al., 1981). Virus aggregation and particle density changes have also been observed in a subset of these viruses (Influenza, Sindbis; duck hepatitis; Edwards et al., 1983; Grgacic and Schaller, 2000; Skehel et al., 1982). However, our sucrose density gradient ultracentrifugation experiments did not reveal density changes in X-MLV following low-pH treatment. Low-pH treatment appears to have a subtle impact on X-MLV, possibly impacting glycoprotein structures, as observed in other viruses. Our observation that capsids still exist after low-pH treatment may explain our observed lack of impact on RNA genomes or RT activity; they are inside the capsids, protected from damage.

It should be noted that we incubated X-MLV at pH 3.8, a harsher pH than that of endosomes (pH ~ 5). This pH is likely to induce greater, more non-specific denaturation of the viruses, which we probably observed as a blurry, mottled appearance of the capsids. A graded pH effect was observed in Sindbis virus where mildly acidic incubation (pH 5.8–6.0) led to enhancement of infectivity, while more acidic conditions (pH ≤ 5.2) led to frank denaturation and loss of infectivity (Meyer et al., 1992).

Our bracket/matrix studies were performed at room temperature for 30 min. Inactivation kinetics of the other viruses was found to be pH and temperature dependent (Bron et al., 1993; Edwards et al., 1983; Stegmann et al., 1987). In addition, product specific variations in the kinetics of X-MLV inactivation have been observed in low-pH unit operations performed below room temperature, at 2–8°C (Abujoub, 2001). Thus, establishment of exposure pH, time, and temperature limits for bracketed generic clearance of X-MLV is warranted. Our bracket/matrix data show that pH ≤ 3.8, time ≥ 30 min, and temperature ≥ 14°C are appropriate for these parameters.

Unlike temperature, time, and pH, it is difficult to imagine how an extraviral protein such as a co-incubated mAb or RP can impact the course of low-pH denaturation of X-MLV capsids and envelope structures, unless they specifically protect the particular glycoproteins. Thus, we propose that the only products that should be exceptions to bracketed generic clearance are those that specifically target retroviral glycoproteins (e.g., an anti-HIV or HTLV-1 mAb). Because cell binding has been shown to protect other types of viruses from low-pH inactivation, we propose that bracketed generic clearance also should not apply where the low-pH incubation occurs in the presence of cells (prior to the initial capture step of recovery).

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